Vanguard + Meridian

Deployment Runbook

From flash to first field mission, with abort conditions

Who this is for. The person who will actually do the work — flashing firmware, running the dock test, taking the boat out. It assumes familiarity with the boat itself but *not* with the Meridian autopilot code. When something surprises you, stop and ask before proceeding.

Estimated total. Seven to ten working days from "we start flashing" to "we have field data in hand." Individual sections range from fifteen minutes (compass calibration) to several days (firmware first-boot). Don't skip § 0.

Contents

Pre-flight status — 19 April 2026

Neural-net position filter:	**WORKING**; three-to-five percent divergence on fine charts in harsh conditions
Landmark fusion (fixed 19 April):	**WORKING**, with a regression test to keep it fixed
Parallel-filter supervisor:	**WORKING** — hot-swaps between filters on divergence
Cold-start, landmark-based:	**WORKING** in simulation (27 m recovered)
Cold-start, blind search:	**WORKING** — returns a top-5 ranked candidate list; operator picks
Cube Plus firmware flash:	**NEEDS BOAT** — the very first step of this runbook
Smart motor-controller wiring:	**NEEDS BOAT** — optional; boat runs without it
Hours in the water on this stack:	zero — § 6 is the first real-hardware contact

1 Before you touch the boat

STEP 1.1 • PRE-FLIGHT CHECKLIST

- ☐ Backup of the *current* autopilot firmware on the Cube Plus saved to a USB stick. **Non-negotiable.** The first boot of new firmware can surprise us, and we need a known-good restore path.
- ☐ Tablet app is paired to the long-range radio on the laptop side. Verify the link by reading heartbeat from the boat while the *old* autopilot is still running. This confirms the radio layer works before we change the firmware underneath it.
- ☐ The Orca Core 2 marine gateway is flowing data on the boat's sensor network. Verify through the Orca's own web interface — depth, satellite position, wind all reading.
- ☐ The seafloor chart tile for the deploy area is saved on the companion computer at `/opt/meridian/charts/<area>.npz`. Fetch commands are in § 10.
- ☐ Companion computer has the `meridian-orca-bridge` service installed **but not started**.

Why all this

Every item above is a failure mode we've actually hit in lab dry-runs. The five minutes you spend here saves four hours of troubleshooting on the dock.

2 Flash Meridian onto the Cube Plus

Estimate: 2 to 4 days. The board configuration file is drafted but has never been compiled. Expect to find missing hardware definitions on the first try.

STEP 2.1 • BUILD THE FIRMWARE IMAGE

```
cd <meridian repo>
./tools/build-board CubeOrangePlus --release
```

STEP 2.2 • BACK UP THE CURRENT AUTOPILOT — DO NOT SKIP

Plug the Cube into the laptop. Pull the existing firmware. Put it on the USB stick you prepared in § 0. If first-boot surprises us, this restores the boat to its known-good state in ten minutes instead of an open-ended debug session.

STEP 2.3 • FLASH IN FIRMWARE-UPLOAD MODE

Hold the BOOT button on the Cube while plugging USB, then:

```
./tools/flash.sh CubeOrangePlus build/out/meridian.bin
```

STEP 2.4 • FIRST BOOT — PROPS PHYSICALLY DISCONNECTED

Watch the serial console. Expected boot sequence:

- Sensor/driver initialization messages
- Sensor probe: motion unit, barometer, compass, SD card
- Position estimator initialization
- Radio heartbeat starts

If boot fails, diff the console output against a known-working old autopilot log. Three most common issues:

- Wrong bus address for the barometer

- Wrong pin mapping for the compass
- SD card controller mismatch — **fatal; must be fixed before proceeding**

3 Load the settings file

Estimate: 30 minutes.

STEP 3.1 • LOAD VIA TABLET

On the tablet: **Tools** → **Parameters** → **Load from file**, then select `vanguard_defaults.meridian-params`. The tablet will set every parameter one at a time. Any parameter the firmware doesn't recognize will appear as a red "UNKNOWN PARAM" line. We expect **zero**.

If you see *any* unknown parameter

Stop. That means the settings file is out of sync with the firmware version — continuing would leave the boat in a half-configured state. Send Jesse the exact parameter name.

Expected total parameters: around 180. Confirm the **verify complete** message before moving on.

4 Bench test

Estimate: 1 hour. Boat on jack stands, props free to spin with physical clearance. Webcam pointed at the rudder so the tablet can watch it move.

STEP 4.1 • ARM AND OPEN BENCH TEST

Arm the boat from the tablet. Confirm dialog. Open the Bench Test overlay from the top-right menu.

STEP 4.2 • RUDDER STEP TEST

Use the slider, or the **STEP** ± 50 pattern. The rudder should follow the commanded position with less than 0.3 seconds of lag.

STEP 4.3 • RUDDER AUTO-TUNE

Hit **AUTOTUNE STEER**. The filter runs a step response, measures the time constant, and suggests controller gains. *Record them — you'll load them in a moment.*

STEP 4.4 • THROTTLE TEST

Step throttle from 0 to 30% for two seconds, then back to 0. Confirm the prop spins and stops cleanly.

5 Compass calibration

Estimate: 15 minutes. Boat on stands, or on a level dock surface with no large iron within three metres.

STEP 5.1 • RUN THE WIZARD

Tablet → **Tools** → **Compass Cal Wizard**. Follow the rotation sequence: nose up, nose down, roll left, roll right, full yaw circle. The wizard writes calibration offsets to the autopilot. Accept the summary dialog.

6 Capture the Orca's wire format

Estimate: 1 day worst case — often ten minutes.

STEP 6.1 • RECORD ONE MINUTE OF TRAFFIC

On the companion computer:

```
sudo tcpdump -i eth0 -w /tmp/orca_capture.pcap
```

Wait 60 seconds with the Orca flowing its normal traffic. `Ctrl-C` to stop.

STEP 6.2 • RUN THE AUTODETECT

```
python -m meridian_orca.protocol.detect /tmp/orca_capture.pcap
```

If the output says **Yacht Devices RAW** or **Signal K**, start the bridge service and move on:

```
systemctl --user start meridian-orca-bridge
```

If autodetect says UNKNOWN

The Orca is speaking a proprietary variant. Send the capture file to Jesse. Expect four to eight hours to add the decoder.

7 Dock test

Estimate: 2 hours. Boat in the water, tied to the slip with at least two lines. The props may engage but cannot meaningfully propel the boat.

STEP 7.1 • START THE BRIDGE AND VERIFY THE SENSOR FEED

Start the bridge service. The tablet should show vessel traffic, depth, wind, satellite position, and compass all flowing.

STEP 7.2 • VERIFY THE DEPTH-CHART FILTER IS LIVE

The status bar reads **NAV: SATELLITE** (green). The depth-chart position marker (a dashed amber B) appears on the map within 30 seconds of the filter getting its first depth samples.

STEP 7.3 • JAMMING-SIMULATION TEST

Press the **GPS Jam** button. Within two seconds:

- Badge flips to amber **DEPTH CHART (SATELLITE JAMMED)**
- If two or more nearby vessels are broadcasting positions within 1.5 km, the badge upgrades to cyan **DEPTH CHART + 2 LANDMARKS** with dashed range lines drawn to each landmark
- The depth-chart marker takes over as the primary position indicator

Press again to restore. Badge returns to green within a few seconds.

STEP 7.4 • DATA-FRESHNESS CHECK

All status-bar indicators — satellite, depth, vessel-broadcast — show green dots.

8 Sheltered-water autonomous mission

Estimate: 2 hours. Open water (sheltered bay or channel), chase boat armed, manual takeover ready on the tablet.

STEP 8.1 • SET WAYPOINTS

Untie the boat. Long-press the map in three or four places to set a waypoint loop.

STEP 8.2 • ENGAGE AUTO

Expected cross-track error: under 5 metres. If worse, check the auto-tune gains from § 3 were actually loaded.

STEP 8.3 • MID-LEG JAMMING DEMO

Fire the jam button while the boat is tracking between waypoints. The boat should continue autonomously using the depth-chart filter for 60 seconds. Cross-track may widen to 20–30 m but the boat should still follow the line. Restore satellite; tight tracking resumes.

STEP 8.4 • RETURN TO LAUNCH

Trigger the return-to-launch command from the tablet. Confirm safe return. Disarm on approach to the dock.

9 Full field deploy

Estimate: 1 day on-site. Repeat § 7 at full demo scale, with video.

STEP 9.1 • CAPTURE ON VIDEO

- Autonomous waypoint loop (green nav)
- Jamming segment with landmark-fusion visible (cyan badge, range lines)
- Collision-avoidance: spawn a simulated target on collision course; watch the evasion
- Station-keeping: drop a virtual station pin; watch the boat hold position

STEP 9.2 • POST-FLIGHT

Pull telemetry logs off the companion computer through the tablet.

10 Abort conditions

- Motion-sensor vibration warning triples. Possible prop imbalance or motor-mount failure.
- Any voltage dip below 11.0 V on a 3-cell pack, or 14.8 V on a 4-cell pack.

- Compass heading drifts more than 30 degrees away from the satellite ground-track direction, sustained for more than five seconds.
- Tablet loses link for more than 20 seconds. The boat *should* return to launch automatically — be ready to recover manually.
- Water inside the hull.
- Props spinning while the boat is marked `DISARMED`. That is a safety-interlock bug. **Kill main power.**

11 Chart fetching — quick reference

STEP 11.1 • PUBLIC US GOVERNMENT SEAFLOOR DATA (TYPICAL DEMO AREA)

```
python -m meridian_orca.chart_loader fetch-etopo \
  --bounds -34.00 -33.90 151.15 151.25 \
  --output /opt/meridian/charts/sydney.npz
```

STEP 11.2 • GLOBAL SEAFLOOR FALLBACK (400 M DETAIL, WORKS ANYWHERE)

```
python -m meridian_orca.chart_loader fetch-gebco \
  --bounds ... \
  --output /opt/meridian/charts/sydney_gebco.npz
```

STEP 11.3 • INSPECT A DOWNLOADED CHART

```
python -m meridian_orca.chart_loader info /opt/meridian/charts/sydney.npz
```

12 Known rough edges — current as of 19 April 2026

- **Neural-net position filter** has a 3-to-5% divergence rate on fine charts in harsh conditions. The supervisor runs the classical filter in parallel and hot-swaps on divergence, which drops end-to-end failure rate to under half a percent.
- **Landmark fusion** had a collapse bug on 18 April (the filter would crush itself to one point and stay stuck). Fixed on 19 April with adaptive noise tolerance plus a regularization boost on fusion steps. A regression test lives in `tests/test_ais_fusion.py`. Validate against your own data before trusting in a new environment.
- **Cold-start without any reference** — no satellite, no nearby landmarks — returns a top-5 ranked list of candidate cells. In feature-poor terrain the right answer often isn't at position #1; an operator picks from the list or a follow-up filter converges on the one whose evidence keeps growing.
- **Smart motor-controller upgrade** is optional. The boat runs on simple pulse-width control without it.
- **Zero hours in the water** on this stack. § 6 is first contact with real hardware.
- **Remote recovery limit.** If the Cube draws power from the boat's bench/main power (not just USB), then cycling the laptop's USB controller *cannot* reset the Cube. We learned this the hard way on 26 April 2026: a Phase-9 stub flashed, the chip went silent on USB, and no software-only recovery worked. Going forward, every Meridian build that touches real hardware must include USB CDC + a reboot-to-bootloader handler before deployment, so the chip is always observable and recoverable from the host. If for any reason a build is deployed without those, recovery requires a physical bench-power cycle by someone at the boat.

13 Getting help

For anything weird, ping Jesse. The fastest path to a fix:

1. `journalctl -u meridian-orca-bridge` — last 200 lines is usually enough
2. Tablet screenshot showing the nav-source badge and status bar at the moment of the issue
3. One sentence: what you expected vs. what happened

Those three things together resolve most issues inside 30 minutes.